

Scattering part 1

Let $\psi(r, \theta, \phi)$ be the Born approximation

$$\psi(r, \theta, \phi) = \frac{e^{ikr}}{r} f(\theta, \phi)$$

where

$$k = \frac{\sqrt{2mE}}{\hbar}$$

and $f(\theta, \phi)$ is the scattering amplitude

$$f(\theta, \phi) = -\frac{m}{2\pi\hbar^2} \int \exp\left(\frac{i\mathbf{p} \cdot \mathbf{x}}{\hbar}\right) V(\mathbf{x}) d\mathbf{x}$$

The density is

$$|\psi|^2 = \psi^* \psi = \frac{|f(\theta, \phi)|^2}{r^2}$$

Let $d\sigma$ be the differential cross section such that

$$d\sigma = |\psi|^2 dA = |\psi|^2 r^2 d\Omega = |f(\theta, \phi)|^2 d\Omega$$

Hence the cross section is completely defined by the scattering amplitude.

$$\frac{d\sigma}{d\Omega} = |f(\theta, \phi)|^2$$

For example, for Coulomb potentials we have the Born approximation

$$\psi(r, \theta, \phi) = \frac{e^{ikr}}{r} \frac{Z\alpha\hbar c}{2E(1 - \cos\theta)}$$

Hence the Rutherford cross section formula

$$\frac{d\sigma}{d\Omega} = \left[\frac{Z\alpha\hbar c}{2E(1 - \cos\theta)} \right]^2$$

Verify dimensions of exponential.

$$\frac{\sqrt{2mE}}{\hbar} r = \frac{\sqrt{[\text{kg}] [\text{J}]}}{[\text{J s}]} [\text{m}] = [1]$$

Verify dimensions of cross section.

$$\left(\frac{Z\alpha\hbar c}{2E} \right)^2 = \left(\frac{[\text{J s}] [\text{m s}^{-1}]}{[\text{J}]} \right)^2 = [\text{m}^2]$$

Eigenmath script